

Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed.

It works by:

- Collecting data
- Training a model
- Making predictions
- Improving performance over time

Types of Machine Learning

1. Supervised Learning

In supervised learning, the model is trained using labeled data.

Examples:

- House price prediction
- Spam email detection
- Student result prediction

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree

- Support Vector Machine

2. Unsupervised Learning

In unsupervised learning, the model works with unlabeled data and finds hidden patterns.

Examples:

- Customer segmentation
- Market basket analysis

Algorithms:

- K-Means Clustering
- Hierarchical Clustering

3. Reinforcement Learning

In reinforcement learning, an agent learns by interacting with the environment and receiving rewards or penalties.

Examples:

- Self-driving cars
- Game AI
- Robot navigation

Components:

- Agent
- Environment
- Reward
- Action

Logistic Regression

Logistic Regression is a supervised learning algorithm used for classification problems. It predicts probability using the Sigmoid function

How It Works:

Instead of predicting a continuous value, logistic regression predicts a probability using the sigmoid function.

Sigmoid Function

- Output ranges between 0 and 1
- Used to map predictions to probabilities
- Decision Boundary
- If probability $\geq 0.5 \rightarrow$ Class 1
- Else $\rightarrow$ Class 0

Applications:

- Disease prediction
- Spam detection
- Customer churn prediction